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Digital and Public Humanities

HOW HISTORIANS USE VISUALIZATION: A CRITICAL READING

ANALYSIS OF THE ARTICLE:
"HOW HISTORIANS USE VISUALIZATION: A CORPUS-BACKED TAXONOMY AND ANALYSIS
FOR CROSS-DISCIPLINARY PRACTICE"

X. CHEN, Y. ZHANG, W. ZHENG, C. MA, X. YUAN

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4,142 articles from 6 journals → 14,021 images → 4,831 visualizations

Mixed methods: corpus analysis + AI labeling + 11 historian interviews

Tool: HiFigAtlas (boundary object)

4 research questions: functions / usage patterns / deficiencies / misuse

INTRODUCTION

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How Historians Use Visualization: A Corpus-Backed Taxonomy and Analysis for Cross-Disciplinary Practice

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Abstract
Visualization in historical research is shifting from isolated attempts to systematic practices. However, data-driven evidence about how historians actually use visualization remains scarce. We present a corpus-driven, mixed-methods study that combines analysis of images from 4,142 research articles across history and digital humanities journals with a collaboratively developed visualization taxonomy and a semi-automatic labeling pipeline. We construct a corpus of 14,021 images, classify 4,831 visualization instances using a hierarchical, domain-informed taxonomy, and analyze patterns of visualization adoption across venues, history subfields, and time. To interpret these patterns, we conduct interviews with 11 historians and use Hi-FigAtlas system as a boundary object to support joint inspection of the corpus. We identify distinct roles for visualizations in historical research: primary-source, evidence-synthesis, communicative, confirmative, and exploratory. We further find that while historians pursue diverse goals with figures, persistent epistemological and practical barriers, such as uncertainty, provenance, justification burden, and publication constraints, impede the adoption of visualization. This work contributes a grounded account of visualization use in historical scholarship and points to opportunities to better support domain-specific needs.

CCS Concepts
• *Human-centered computing* → *Visualization theory, concepts and paradigms*; • *Applied computing* → *Arts and humanities*;

1. Introduction
With the rise of digital humanities (DH) and quantitative history, historians are increasingly using data-driven approaches [Gra22, Dru11]. As the need to manage and interpret data grows, visualization is increasingly used in historical research for communication, analysis, and knowledge production. However, a systematic understanding of how historians use visualization is still lacking.

The current understanding of visualization usage in historical scholarship faces two limitations. First, existing literature and design studies predominantly focus on digital humanities [PLP⁺23, JFCS17, BEC⁺18]. While valuable, they represent a technically oriented subset of the field, often overlooking the practices within “traditional” history journals where the majority of domain knowledge is produced. Furthermore, these studies rarely characterize the diversity of visualization styles and functions in research articles

age by historians, the landscape remains unclear. Consequently, it is difficult to answer the following questions at scale: Which visualization techniques are effectively adopted in historical research? How do visualization types distribute over time, publication venues, and history subfields? Why do historians use visualization, and what are their underlying motivations? Which visual forms are underused, misused, or largely ignored in historical practice? What are the potential barriers hindering their adoption?

To address these gaps, we conduct a corpus-driven mixed-methods study of visualization in historical research articles. First, we construct a three-level corpus from research articles in general history and digital humanities journals, linking articles, extracted images, and visualization instances. Second, we collaboratively develop a domain-informed visualization taxonomy with historians and scale labels to the full corpus with a semi-automatic pipeline.

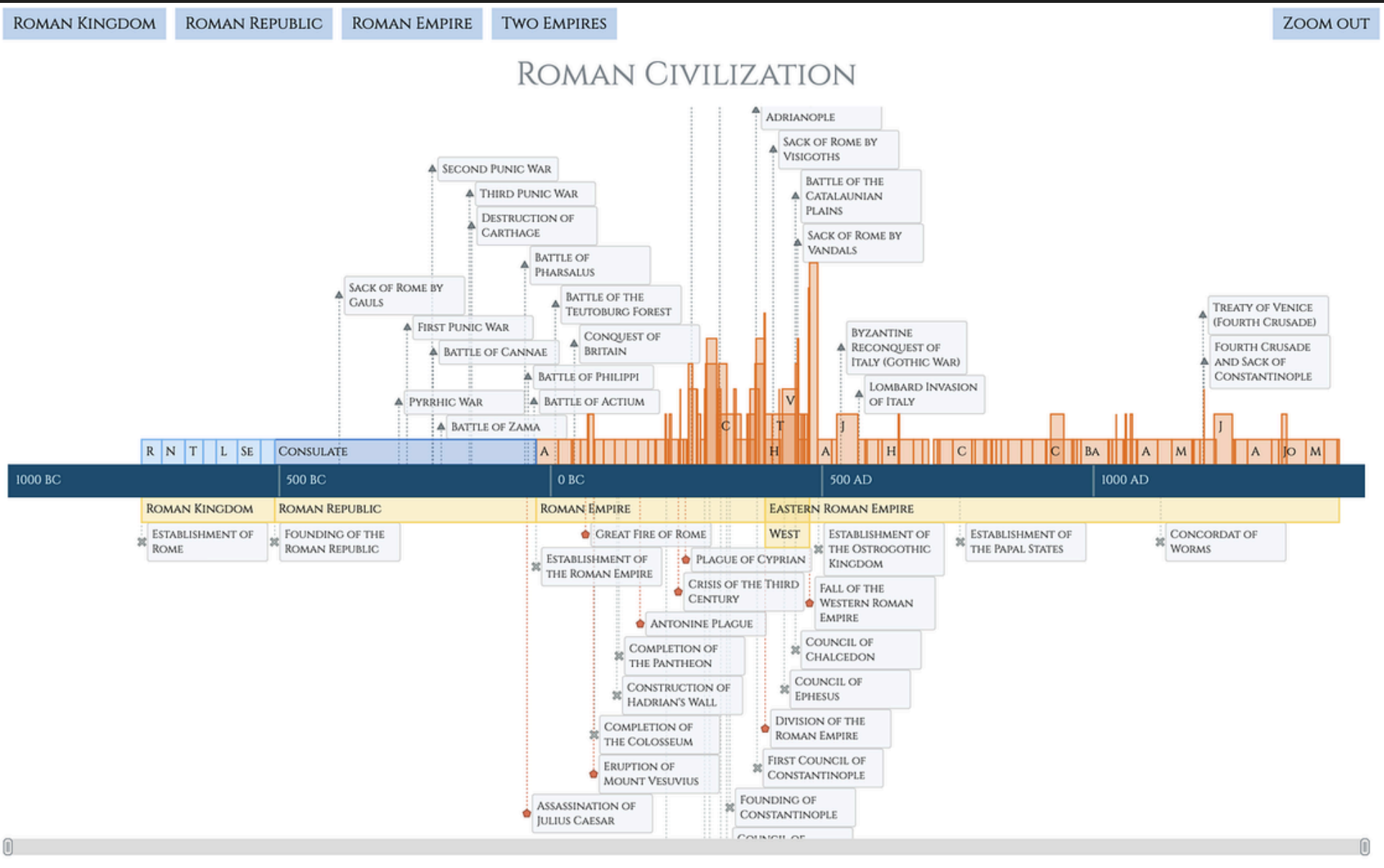
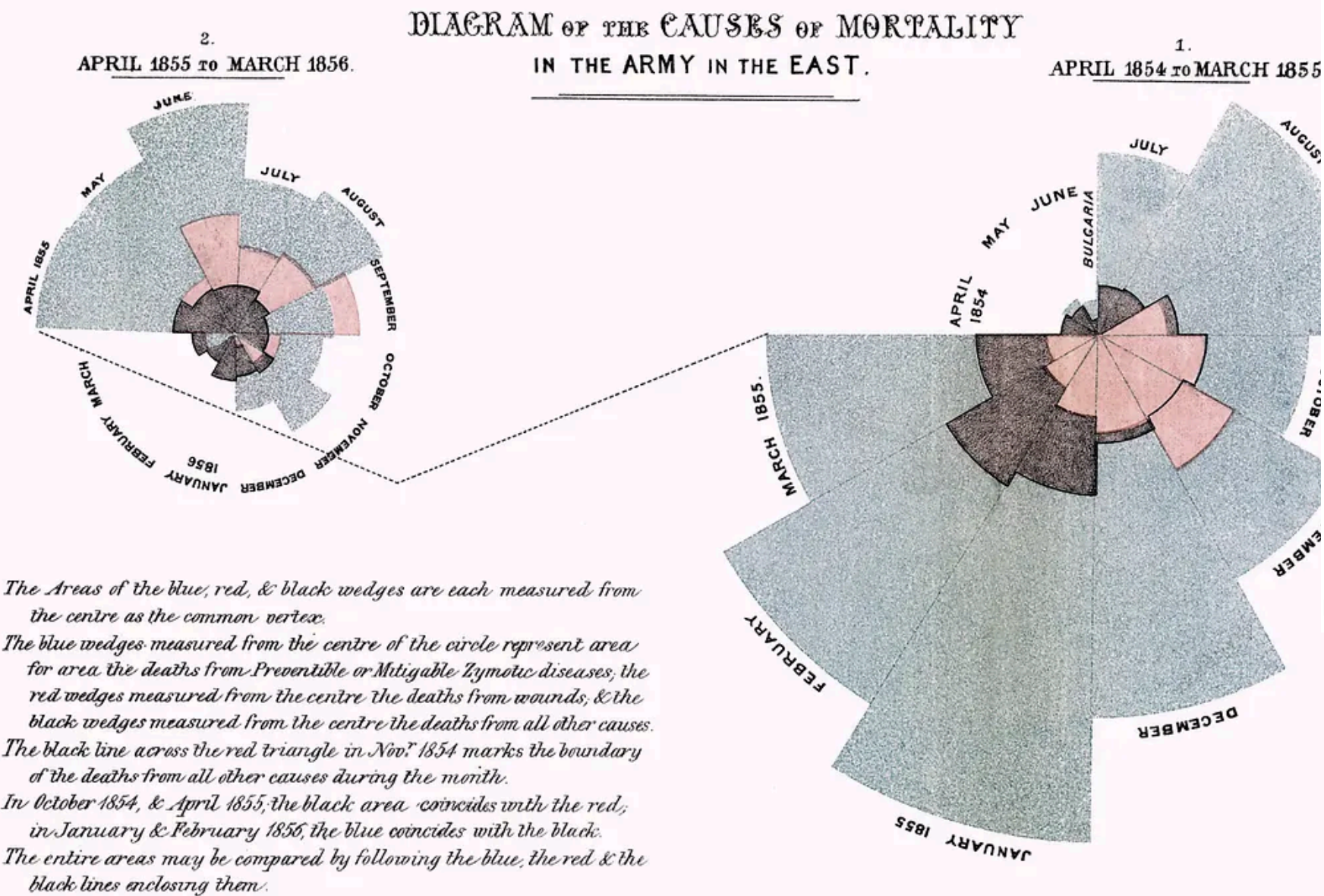
A TAXONOMY OF HISTORICAL VISUALIZATIONS

Table 1: *Research articles per journal in the final corpus.*

Journal	Articles	Share
Past & Present	1,403	33.87%
The American Historical Review	1,328	32.06%
Historical Methods	598	14.44%
History and Technology	322	7.77%
Digital Scholarship in the Humanities	252	6.08%
Digital Humanities Quarterly	239	5.77%
Total	4,142	100.00%

- 47 categories of visualization types, organised in a hierarchy
- Most used: Maps (33.93%), Line charts (23.95%), Bar charts (13.08%)
- **Timelines and storylines nearly absent** – surprising given that time is central to history
- Visualization is growing: found in 1 in 5 articles in 1983, now in nearly 1 in 2 (2024)
- DH journals: up to 84% of articles include visualization
- Traditional history journals: below 30%
- Economic history = most visualization-intensive subfield

ROLES OF VISUALIZATION IN HISTORICAL RESEARCH

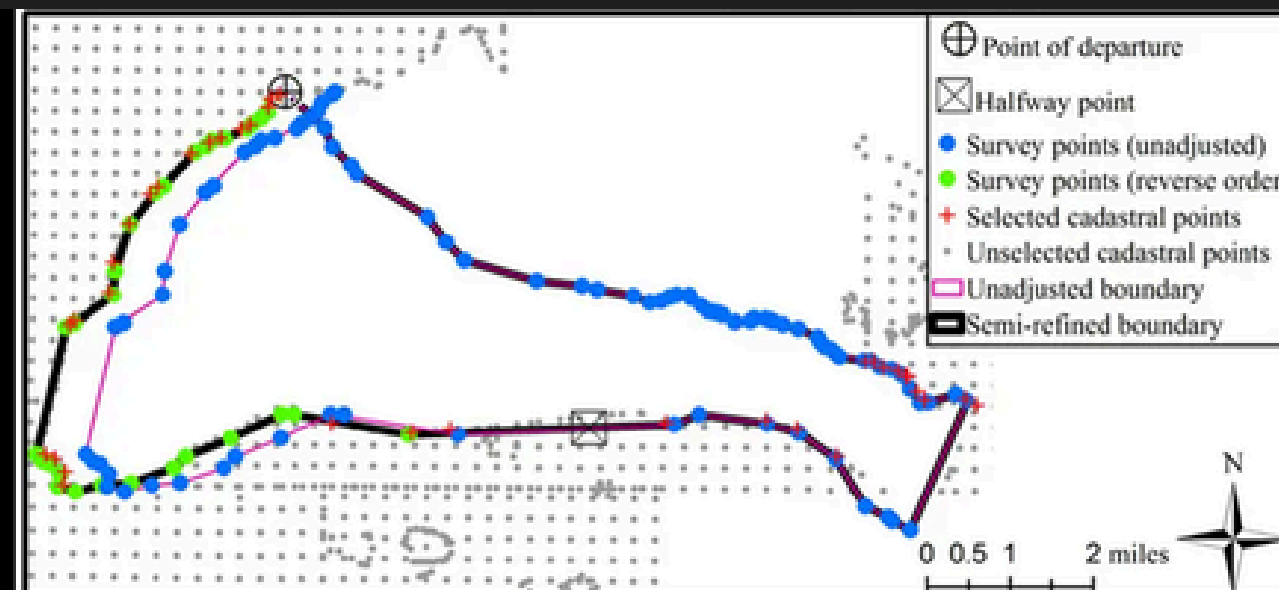


- 1) Primary-Source – historical artefact as evidence
- 2) Evidence-Synthesis – organising scattered sources
- 3) Communicative – presenting conclusions (most common)
- 4) Confirmative – modelling uncertainty, testing hypotheses
- 5) Exploratory – pattern discovery through interactive tools

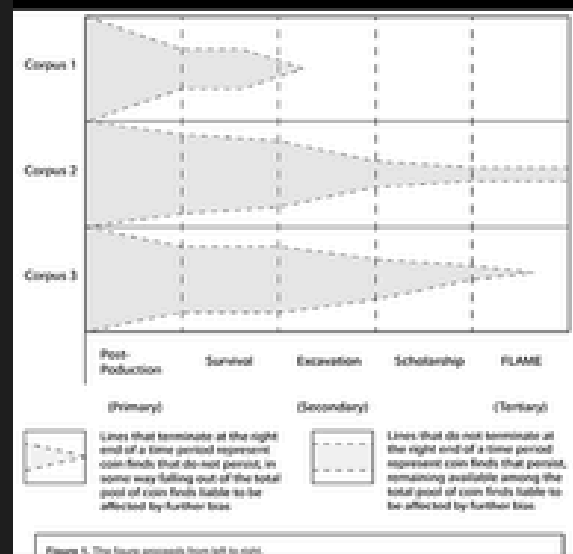
- Primary-source → Buchholz's 1956 epidemic map (Black Death)
- Evidence-synthesis → family trees, tables (used by all 11 interviewees)
- Communicative → 404 flowcharts in corpus
- Confirmative → Fogel's scatter plots on slave prices
- Exploratory → TOFLIT18 network + data querying



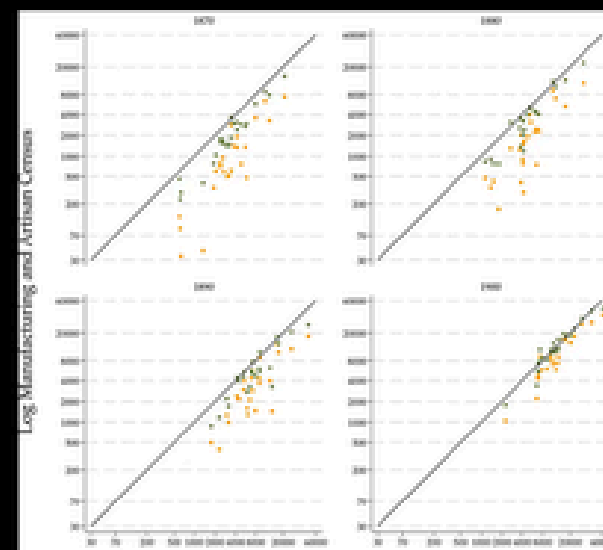
map flow map



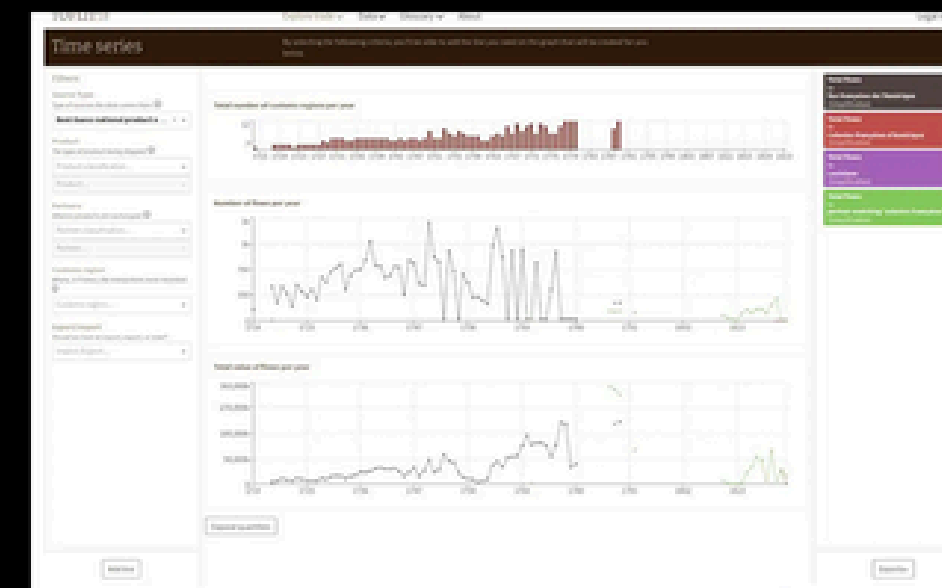
map route map



flowchart



scatter plot simple scatter plot
small multiples



line chart grouped line chart

WHY HISTORIANS VISUALIZE: MOTIVATIONS AND BENEFITS

- Managing scale – compress large archives without losing accountability
- Externalising relations – make networks, lineages, pathways revisitable

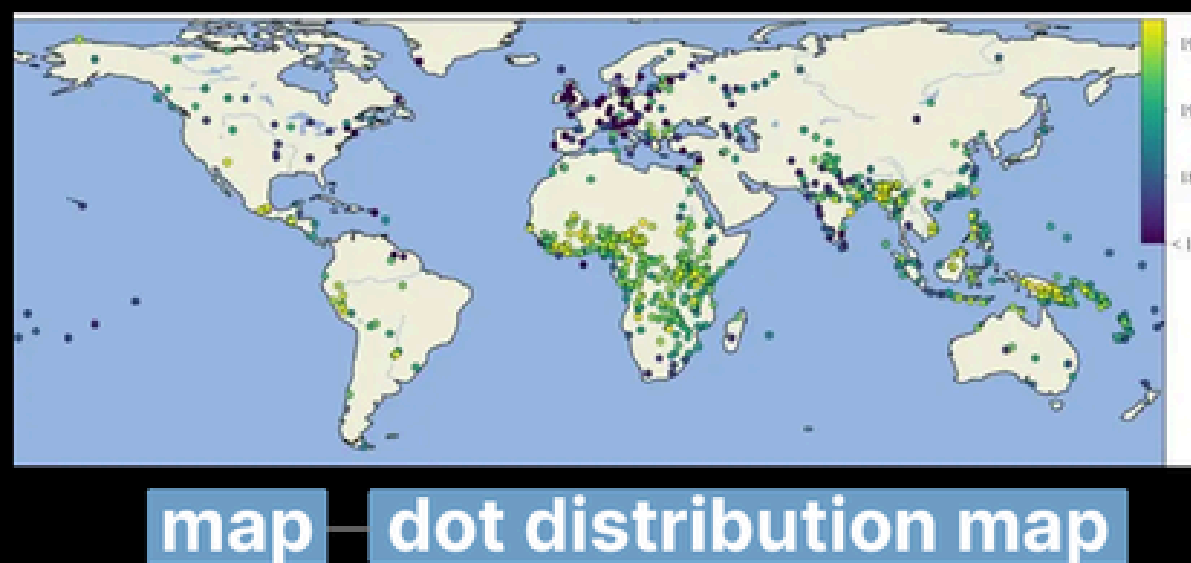
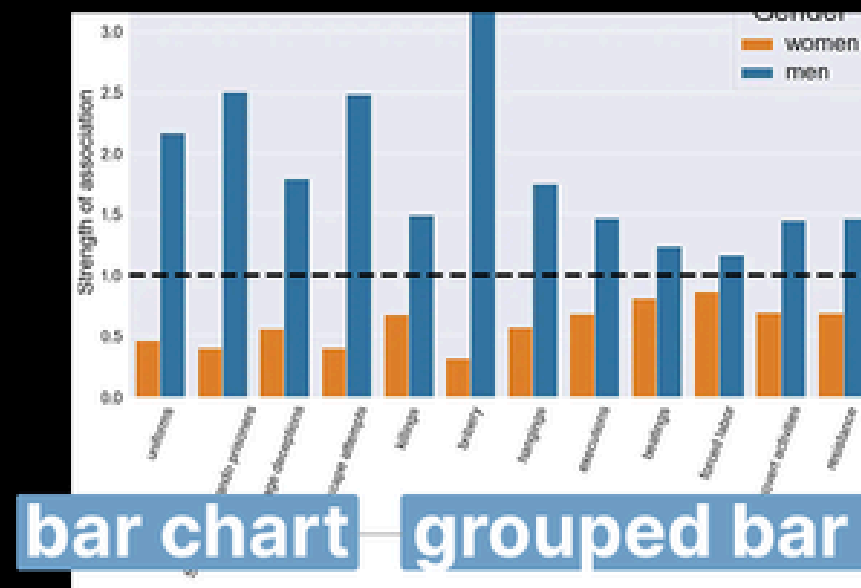
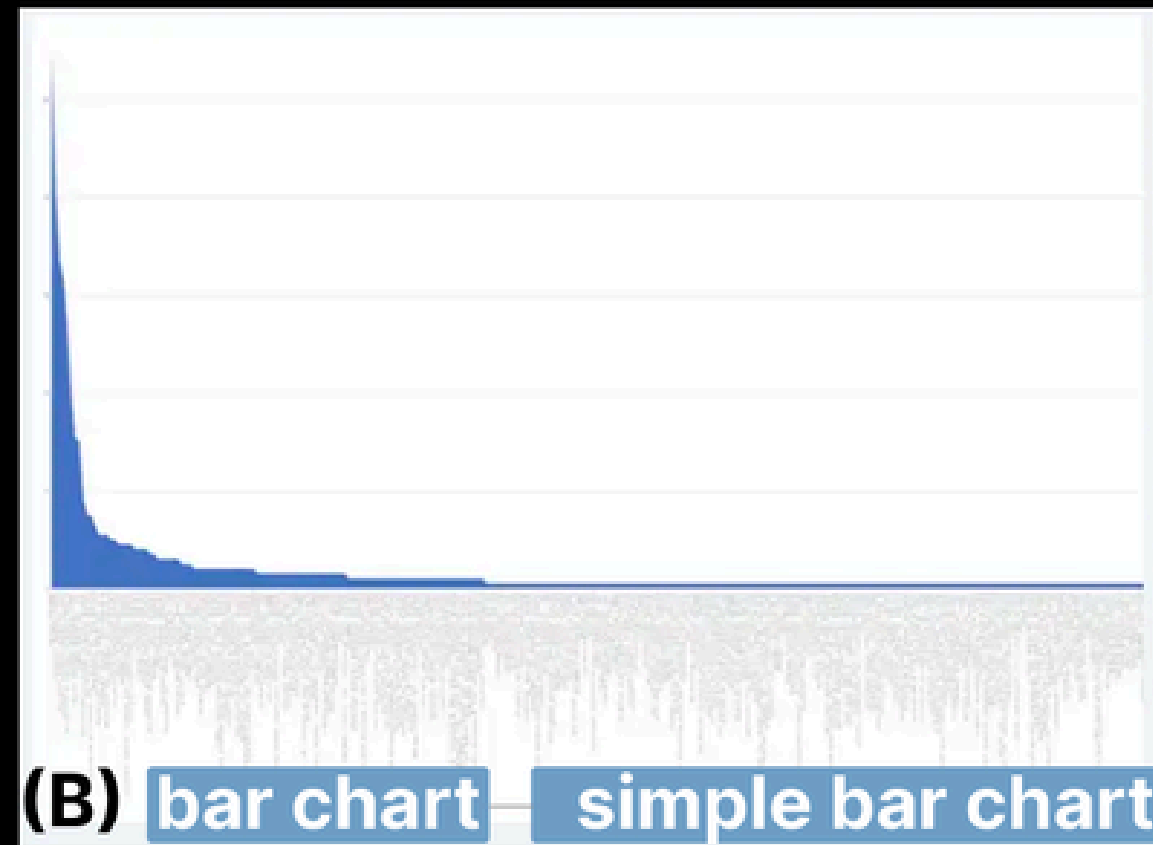
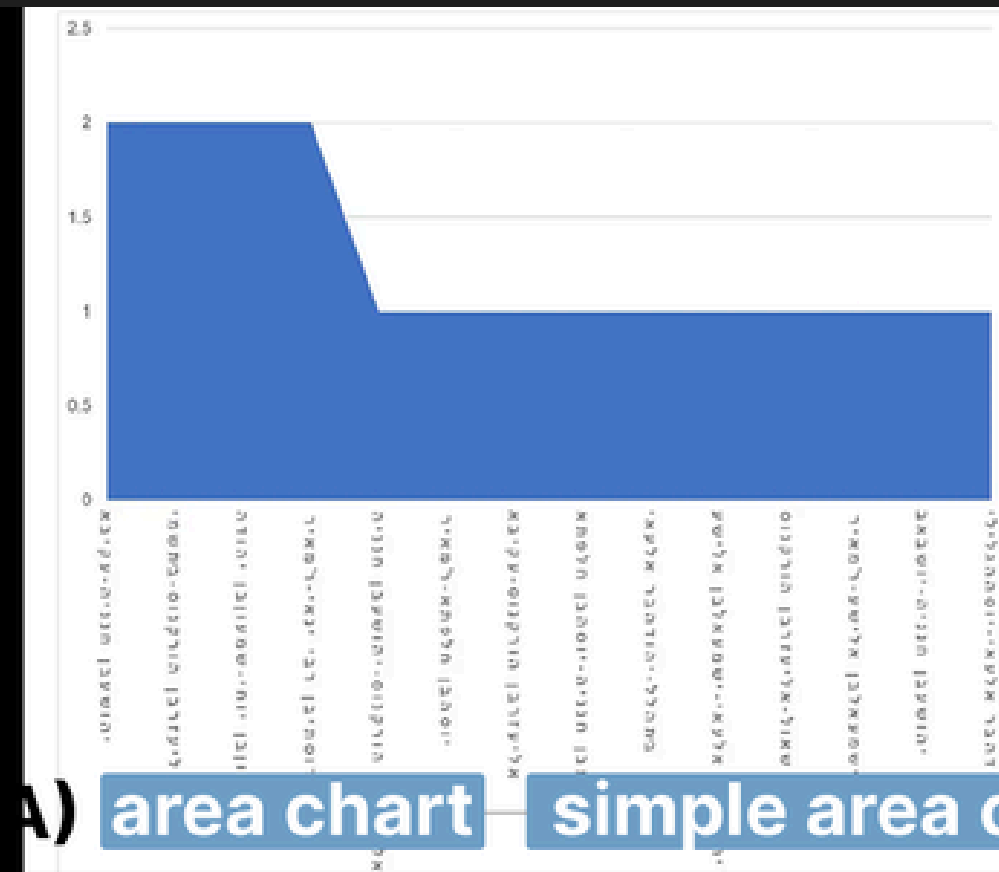


- Expressiveness – give shape to abstract arguments
- Methodological legitimacy – provenance and transparency

- Epistemological – images conceal subjectivity; rhetorical authority; time resists linearity
- Data – algorithmic choices need justification; reduction loses meaning
- Practical – steep learning curves; literacy gaps; print constraints

BARRIERS AND CONCERNS

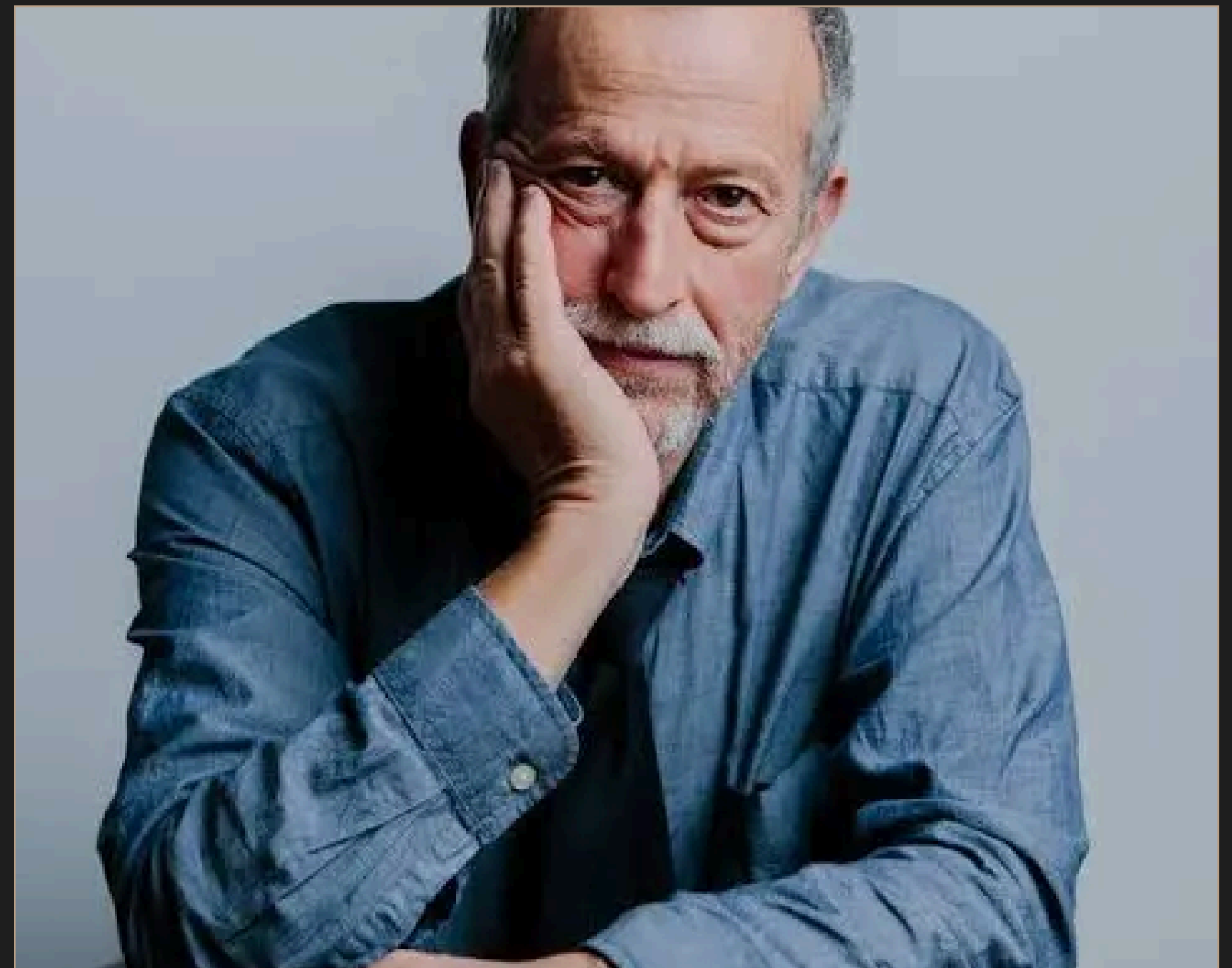
MISUSE PATTERNS



- (A) Area chart used for categories – implies a trend that isn't there
- (B) Too many labels – becomes unreadable
- (C) Gender reduced to Male/Female – erases complexity
- (D) Color used to show time – makes comparison impossible

THROUGH THE LENS OF THE COURSE

The five functional roles show that visualization is not one thing – it can serve completely different purposes depending on the research stage. This connects directly to the course's argument that a model is always built for a specific purpose, and that purpose shapes what gets included and what gets left out. The study itself is a corpus analysis, raising the same selection-bias questions discussed in class. Like Moretti's distant reading, its quantitative patterns still require interpretation. And the 47-taxon taxonomy is an act of operationalisation – transforming an abstract concept into measurable categories, at a specific granularity that balances detail against the cost of annotation.



REFLECTION

Historians are not afraid of technology – they are careful about evidence. Every visualisation involves choices: what to include, how to categorise, and which scale to use. These choices shape the argument, often invisibly. A chart can make uncertain data look settled, a map can impose modern borders on ancient spaces, and a binary category can erase historical complexity. Historians are trained to be suspicious of exactly this kind of silent simplification. So their resistance is not backwardness – it is a form of **epistemological discipline**, the same rigour they apply to any other source.